Model Selection Note

SME Procurement Preparation Agent

BSE4104 - Week 2

Selected Model: GPT-4o-mini OpenAI

Rationale for Selection

- Capability. Strong instruction following, good structured output tables and reliable reasoning for simple analytical tasks such as comparing stock levels against reorder points.
- Cost. Significantly cheaper than GPT-4o or Claude 3.5 Sonnet, making it suitable for frequent testing and iteration during the 8 week project.
- Latency. Fast response times important for a responsive baseline demonstration.
- Privacy and access. Accessible via OpenAI API with a standard API key. No fine tuning required. Data sent is only synthetic inventory information.
- Ease of integration. Excellent Python SDK support and simple chat completions API allowing rapid development of the baseline.

Alternatives Considered

Model	Reason for rejection
GPT-4o	Higher cost for limited extra benefit at this stage.
Claude 3.5 Sonnet	Slightly better reasoning but higher cost and slower.
Grok	Very fast and cheap but weaker structured output consistency.

Limitations Accepted

- The model can still occasionally hallucinate if the prompt is weak (mitigated by strict prompt constraints).
- No native tool calling used yet (will be added in Week 4).
- Context window is more than sufficient for our small synthetic datasets.

Decision

GPT-4o-mini offers the best balance of cost, speed, reliability, and ease of use for Week 2 baseline development. We will re evaluate after Week 4 if stronger reasoning or better tool calling is required.